

SHORYA GUPTA

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shoryagg7.github.io/portfolio · GitHub · LinkedIn · Codeforces · CodeChef · LeetCode

EDUCATION

Thapar Institute of Engineering and Technology

Aug 2023 – Jul 2027

Bachelor of Engineering, Computer Science Engineering

CGPA: 8.87 / 10.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Parallel Computing, Software Engineering

ACHIEVEMENTS

- Codeforces Expert (Max Rating: 1687) across 16 rated contests
- LeetCode peak contest rating 2014 over 23 rated contests; 844 problems solved, 573 of them Medium or Hard
- CodeChef 3-Star (Max Rating: 1733)
- Solved 2000+ DSA and competitive programming problems across Codeforces, CodeChef, LeetCode, and Geeks-forGeeks
- Runner-Up, Inter-College Coding Contest (100+ participants)
- Merit Scholarship, Thapar Institute — GPA-based award for academic excellence (9.34 AGPA)

SKILLS

Languages: C++ (Intermediate), Python, SQL

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, Asynchronous Programming

Databases & Caching: PostgreSQL, Redis, Vector Databases (FAISS)

Messaging & Streaming: Apache Kafka, Event-Driven Architecture, Pub/Sub

Systems: System Design, Rate Limiting, Caching, Idempotency, Fault Tolerance, Concurrency Control

DevOps & Observability: Docker, Docker Compose, Git, GitHub Actions, CI/CD, Prometheus, Grafana, Linux

AI / GenAI: LLM Integration, Retrieval-Augmented Generation (RAG), Embeddings, Prompt Engineering

TECHNICAL PROJECTS

DeliverIQ – Distributed Order Dispatch API [GitHub]

2026

Python, FastAPI, PostgreSQL, Redis, Kafka, Docker, Prometheus, Grafana

- Built a horizontally scaled dispatch service (3 API replicas) using two-phase claiming with PostgreSQL `SELECT FOR UPDATE SKIP LOCKED`; diagnosed and fixed a double-dispatch race, verifying zero duplicate assignments under concurrent load
- Designed the dispatch core: a priority-queue scheduler ($O(\log n)$) with time-based aging to prevent starvation, and geohash-based rider matching ($O(1)$ cell lookup vs $O(n)$ scan) with fairness-banded assignment that balances rider earnings within delivery SLA
- Streamed order events through Apache Kafka to independent notification, analytics, and audit consumer groups — durable, replayable delivery replacing fire-and-forget Pub/Sub
- Hardened for production: Redis token-bucket rate limiting, idempotency keys for safe retries, JWT-protected admin endpoints, and a validated order state machine; load-tested with Locust at ~ 123 RPS, p99 220 ms
- Instrumented with Prometheus/Grafana dashboards, structured JSON logs with request-ID correlation, and DB/Redis/Kafka health checks; Dockerized and deployed to Railway with GitHub Actions CI

SemanticCache – LLM Caching & RAG Gateway [GitHub]

2026

Python, FastAPI, Redis, FAISS, LLM APIs, Docker

- Built an API gateway that semantically caches LLM responses — embeds queries and serves cached answers for similar requests, cutting redundant LLM calls, cost, and latency
- Implemented a RAG pipeline over a document corpus with vector embeddings and FAISS similarity search for grounded answers
- Tuned the cache-hit similarity threshold against labeled query pairs to trade hit rate against false hits, since a wrong cached answer costs more than a cache miss
- Added per-user rate limiting and token-cost tracking in Redis to control LLM spend and prevent abuse

EXTRACURRICULAR ACTIVITIES

- Vice Head, Photography, FAPS Society (Fine Arts & Photography Society), Thapar Institute
- State-level football player (2022); long jump athlete, college athletics team, at inter-college competitions